

Multimodal Evaluation of Bidirectional Human–AI Alignment

Dr. Bhagyabati Moharana
b.moharana@research.ait.ie
Technological University of the
Shannon
Athlone, Ireland

Dr. Eoghan Hynes
eoghan.hynes@tus.ie
Technological University of the
Shannon
Athlone, Ireland

Dr. Bryan Dunphy
bryan.dunphy@tus.ie
Technological University of the
Shannon
Athlone, Ireland

Dr. Conor Keighrey
conor.keighrey@tus.ie
Technological University of the
Shannon
Athlone, Ireland

Dr. Niall Murray
nmurray@research.ait.ie
Technological University of the
Shannon
Athlone, Ireland

Abstract

As artificial intelligence systems increasingly function as conversational and embodied partners, alignment should be understood as an interactional process rather than only a model-level property. This paper presents a multimodal framework for operationalising bidirectional human–AI alignment across conversational coordination, behavioural synchrony, physiological modulation, and perceived alignment. The framework is examined in immersive XR interactions between 62 participants and an AI-driven character. Alignment was assessed using conversational metrics derived from voice conversations, positional metrics extracted from HMD heatlogs and video tracking, physiological indicators including inter-beat interval (IBI), and subjective metrics capturing perceived overlap, emotional engagement, and overall experience. Results show that positional synchrony forms a coherent embodied alignment construct and is associated with physiological modulation. Perceived overlap strongly predicts emotional engagement and overall experience. These findings support a multimodal approach to evaluating human–AI alignment in interactive systems.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing design and evaluation methods.**

Keywords

Human–AI Alignment; Bidirectional Interaction; Embodied Interaction; Conversational Synchrony; Multimodal Evaluation; Physiological Synchrony; Extended Reality (XR); Affective Computing

ACM Reference Format:

Dr. Bhagyabati Moharana, Dr. Eoghan Hynes, Dr. Bryan Dunphy, Dr. Conor Keighrey, and Dr. Niall Murray. 2026. Multimodal Evaluation of Bidirectional

Human–AI Alignment. In *Proceedings of (BiAlign @ CHI 2026)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Artificial intelligence systems are increasingly embedded in social and collaborative settings [1, 46]. Rather than functioning only as task-based tools, AI systems now participate in dialogue, shared decision-making, and immersive interaction [12, 21]. In these contexts, alignment is no longer limited to output correctness or safety constraints. It also concerns how humans and AI adapt to each other during interaction [28, 31, 40].

Research in human–human interaction shows that behavioural and physiological synchrony are associated with successful collaboration, shared understanding, and engagement [4, 16, 35]. Conversational coordination, gesture alignment, and autonomic coupling have been linked to improved interaction outcomes [13, 35]. However, this symmetry does not directly transfer to human–AI collaboration. AI systems do not possess biological physiology, and therefore, classical models of interpersonal physiological synchrony cannot be fully applied [36]. This creates a fundamental challenge: how should alignment be evaluated when reciprocal physiological coupling is inherently asymmetric?

At the same time, assessing User Experience (UX) in human–AI collaboration remains difficult [15]. Quality of User Experience (QoE) refers to the overall subjective and experiential quality of interaction, including perceived engagement, presence, emotional response, and satisfaction [58]. In collaborative AI settings, QoE emerges from complex interactional dynamics rather than from task performance alone. Single-modality metrics are insufficient to capture these multidimensional processes. Current alignment processes often focus on model-level value alignment or reinforcement-based control [10, 17]. However, when AI systems participate in immersive and embodied interaction, alignment must also be understood as a lived, relational phenomenon [12, 49]. In these settings, humans and AI do not merely exchange information; they coordinate actions, synchronise rhythms, negotiate turn-taking, and influence each other’s physiological and emotional states [6, 22, 40]. Evaluating alignment in such systems, therefore, requires multimodal approaches that capture reciprocal adaptation across subjective, behavioural, and physiological layers [14, 21].

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BiAlign @ CHI 2026, Barcelona, Spain

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ACM ISBN 978-1-4503-XXXX-X/2026/04
<https://doi.org/XXXXXXX.XXXXXXX>

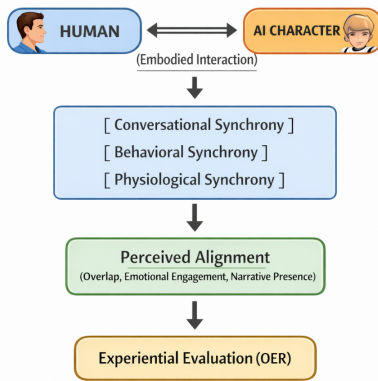


Figure 1: The proposed framework of human-AI embodied interaction. Multimodal synchrony mechanisms operate as objective coupling processes that influence users’ perceived alignment with the AI character, which subsequently shapes overall experiential evaluation.

To address this problem, this paper proposes a structured four-layer framework for operationalising bidirectional human-AI alignment in embodied interaction (Figure 1). The framework evaluates alignment across:

- (1) Conversational coordination (voice-based timing and overlap),
- (2) Behavioural synchrony (positional alignment derived from heatlogs and video tracking),
- (3) Physiological modulation (cardiac inter-beat interval dynamics),
- (4) Perceived alignment (subjective overlap and engagement measures).

Rather than seeking direct physiological synchrony between human and AI, the framework examines how interactional coordination shapes human behavioural coherence, autonomic responses, and subjective QoE. Experiential outcomes are positioned as the result of cross-layer alignment processes.

The framework is empirically grounded in immersive XR-based human-AI interaction. By integrating conversational, positional, physiological, and subjective metrics, this work contributes a multimodal methodology for evaluating bidirectional alignment in collaborative AI systems.

The study addresses the following research questions:

- **RQ1:** How can bidirectional human-AI alignment be operationalised across conversational, behavioural, physiological, and subjective layers?
 - **RQ2:** How do these alignment layers relate to human QoE in collaborative AI interaction?
 - **RQ3:** What implications do multimodal alignment dynamics have for evaluating and designing adaptive human-AI systems?
- By reframing alignment as a measurable, multilayered interactional process and situating QoE as a central outcome, this work contributes a structured evaluation approach to ongoing discussions on bidirectional human-AI alignment.

2 A Multimodal Evaluation Framework for Bidirectional Human-AI Alignment

Research in human-human collaboration shows that behavioural and physiological synchrony are associated with engagement and cooperative success [4, 23, 35]. Parallel work in human-AI interaction has emphasised value alignment, system transparency, and interaction design mechanisms [1, 45]. However, few studies have proposed structured multimodal methodologies for evaluating bidirectional alignment in collaborative AI settings, particularly under physiological asymmetry [44, 51].

Building on synchrony research while addressing its limitations in human-AI contexts, this section introduces a four-layer evaluation framework for operationalising bidirectional alignment as shown in (Figure 1). Each layer captures a distinct yet interrelated dimension of reciprocal adaptation across interaction.

2.1 Conversational Alignment

The first layer concerns temporal coordination in dialogue. Conversational alignment reflects how humans and AI negotiate turn-taking, interruptions, overlap, pacing, and response latency[11]. In human-human interaction, conversational synchrony is widely recognised as a marker of rapport, cooperation, and shared attention [32]. In human-AI interaction, similar dynamics may signal reciprocal adaptation rather than one-sided command-response exchange [8, 43]. Conversational alignment is operationalised through audio-derived interaction metrics, including speech overlap and timing patterns between the participant and the AI character. These measures capture whether interaction unfolds as a rigid sequence or as a dynamically negotiated exchange [29]. Importantly, conversational alignment is inherently bidirectional: both human and AI contribute to the temporal structure of dialogue [24, 32].

2.2 Behavioral Alignment

The second layer addresses embodied synchrony. Alignment in embodied AI systems extends beyond linguistic exchange. Humans coordinate posture, gesture, head orientation, and spatial positioning in ways that reflect shared attention and mutual attunement [12, 26]. When interacting with embodied AI agents, including virtual avatars, robotic systems, or XR characters, such behavioural coupling becomes measurable [41, 50].

Behavioural alignment is operationalised using MediaPipe-based pose tracking of the human participant [27, 40, 55]. Heatlogs from the HMD also provided the position of head and hands of both human and AI character. From tracked head and wrist keypoints, the following measures are derived:

- Head movement synchrony
- Wrist (gesture) synchrony
- Composite synchrony indices

These metrics quantify coherence and coordination over time, enabling analysis of whether movement patterns converge, diverge, or exhibit coordinated variability. Behavioural alignment thus reflects embodied coupling beyond verbal interaction.

2.3 Physiological Co-Regulation

The third layer concerns autonomic alignment. In human-human interaction, synchrony in physiological signals such as heart rate

and inter-beat interval (IBI) has been associated with emotional attunement and shared affective states [33, 57]. In human-AI contexts, physiological responses may indicate whether embodied alignment extends into autonomic regulation [36].

Physiological indicators, particularly inter-beat interval (IBI), are examined to determine whether behavioural and conversational coordination co-occur with shifts in arousal [5, 9, 18]. Rather than treating physiological activation solely as an outcome variable, it is considered part of the alignment process itself, capturing how interaction modulates autonomic states.

This layer bridges technical coordination with affective embodiment. If conversational and behavioural alignment coincide with systematic physiological modulation, alignment may reflect deeper co-regulation rather than surface-level interactional coupling.

2.4 Perceived Alignment

The fourth layer concerns subjective alignment, or how interactional reciprocity is experienced by participants. Technical synchrony does not necessarily imply perceived attunement; conversely, participants may report a strong sense of alignment even when objective coordination varies [20, 42, 53].

Perceived alignment is operationalised using a self-report overlap score capturing participants' sense of conversational reciprocity, timing balance, and mutual responsiveness with the AI character [30]. Additional experiential measures, including emotional engagement and narrative presence, provide context for understanding how perceived alignment relates to broader interactional quality [52, 54].

Distinguishing perceived alignment from embodied alignment enables examination of whether subjective impressions accurately reflect measurable multimodal coordination [38].

2.5 Interactions Across Layers

Although analytically separable, conversational, behavioural, physiological, and perceived alignment are not independent. Alignment may emerge when coordination across multiple layers converges. For example, conversational timing may facilitate behavioural synchrony; behavioural synchrony may co-occur with physiological arousal patterns; and perceived alignment may reflect, though not fully capture, these embodied processes.

This layered model enables a comprehensive evaluation of bidirectional alignment across multiple modalities. It supports a shift from static alignment metrics toward dynamic, multimodal alignment states that evolve throughout interaction.

Subsequent sections ground this framework in empirical observations from embodied human-AI interaction, demonstrating how each layer can be operationalised and how cross-layer dynamics contribute to reciprocal alignment.

3 Empirical Context and Study Design

This study was designed to empirically ground the proposed bidirectional alignment framework within an embodied human-AI interaction setting. A within-subject, single-group experimental design was conducted in a dedicated immersive XR environment. The methodological objective was to capture alignment as a distributed and multimodal phenomenon by integrating validated self-report



Figure 2: (Top) Physical immersive theatre space where the participant wears a mixed-reality HMD. (Bottom) Virtual environment displaying the AI character (Chris) with whom the participant interacts during the XR narrative session.

measures with conversational, physiological, and behavioural indicators recorded during interaction.

3.1 Interaction Environment and AI System

The study was conducted within *Cadence of Altered Illusions* (CAI), an immersive XR experience staged in a controlled performance environment. Participants interacted with the system via an Oculus HMD, enabling six degrees of freedom (6DoF) embodied movement [3]. This configuration supported natural head orientation, spatial positioning, and gesture expression, creating conditions in which behavioural synchrony between participant and AI avatar could be meaningfully captured [19]. As illustrated in Figure 2, the top panel depicts the physical immersive theatre space in which the participant wears a mixed-reality head-mounted display (HMD) and moves freely within the performance area. The bottom panel presents the corresponding virtual environment, where the AI character Chris appears and engages the participant during the XR narrative session.

This tight coupling between the physical and virtual layers enabled simultaneous capture of multimodal interaction data. While the participant navigated the physical space, interaction unfolded within the rendered XR scene, allowing synchronized recording of HMD-derived spatial heat logs, conversational audio, outward video footage of embodied movement, and physiological signals for alignment analysis.

The HMD heat logs provided continuous six degrees of freedom (6DoF) positional and rotational data, enabling precise reconstruction of the participant's head orientation and spatial trajectory, as well as the AI character's relative position within the shared virtual environment. Outward-facing video recordings added contextual detail regarding full-body movement dynamics, while the recorded dialogue between the participant and Chris supported conversational synchrony analysis.

At the centre of the interaction was an AI-driven conversational character embodied through a customised large language model.

The AI functioned as an adaptive interaction partner rather than a scripted non-player character. Customisation included:

- (1) **Instructional role definition:** The model embodied the character across distinct narrative stages, modulating tone, affect, and contextual knowledge.
- (2) **Contextual grounding:** Narrative documentation and design materials ensured consistency of references and interactional continuity.
- (3) **Interactive responsiveness:** Open-ended dialogue enabled participant utterances to influence conversational timing, emotional tone, and environmental transitions.

This configuration allowed conversational timing, overlap, and affective exchange to emerge dynamically, creating conditions suitable for examining reciprocal alignment processes.

3.2 Participants

A total of 62 participants took part in the study. The mean age was 41 years ($SD = 10$). The sample was male-skewed ($n = 40$), with 92% reporting higher education and prior familiarity with VR technology. Each participant engaged individually with the AI character, enabling analysis at the level of the human-AI dyad. All data were analysed as a unified cohort.

3.3 Evaluation Protocol

The protocol consisted of four primary stages:

- (1) **Consent and Preparation:** Participants provided informed consent and were fitted with physiological sensors.
- (2) **Sensor Configuration:** A Polar H10™ chest strap and Emotibit™ sensor were attached to capture cardiac and physiological signals. Separate microphones recorded the participant and AI speech.
- (3) **XR Interaction:** The immersive interaction lasted approximately 52 minutes. Multimodal signals were recorded continuously.
- (4) **Post-Interaction Measures:** Participants completed standardized self-report questionnaires.

The protocol was designed to preserve immersion while enabling continuous multimodal data capture.

3.4 Multimodal Data Acquisition

Data collection was structured to operationalise the four-layer alignment framework. Raw multimodal signals were recorded continuously during interaction and synchronised for cross-modal analysis. Table 1 summarises the raw data streams and derived metrics used to quantify alignment across layers.

Conversational Data. A single audio recording captured both participant and AI speech during interaction. Post-processing included segmentation to distinguish speakers. Conversational timing metrics, including turn-taking intervals, speech overlap duration, and pacing dynamics, were derived to quantify temporal coordination [28, 32].

Behavioural Data. Embodied alignment was quantified using two complementary data sources. First, head pose trajectories were derived from HMD heat logs. The Oculus HMD provided continuous 6DoF positional and rotational data, enabling precise estimation of

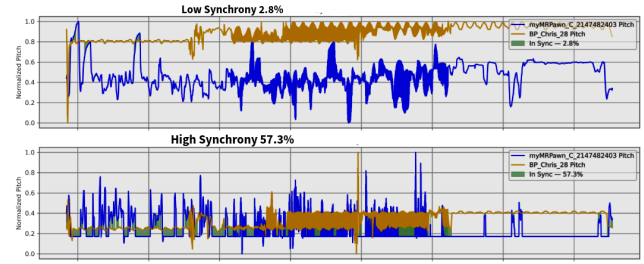


Figure 3: Representative examples of head pitch synchrony derived from HMD heat logs. Blue and orange lines represent human and AI head pitch trajectories, respectively. Green shaded regions indicate time windows classified as synchronised based on pitch alignment criteria. The examples illustrate low (2.8%) and high (57.3%) synchrony segments, demonstrating the operationalisation of behavioural alignment over time.

both the participant’s head orientation and the spatial position of the AI character (Chris) within the XR environment. Head pitch trajectories were extracted and temporally aligned between the human participant and the AI avatar. Behavioural synchrony was computed using sliding-window coherence analysis applied to these pitch signals, resulting in synchrony percentages reflecting the proportion of temporally aligned head dynamics. As shown in Figure 3, low synchrony segments (2.8%) exhibit largely divergent pitch trajectories, whereas high synchrony segments (57.3%) demonstrate sustained temporal alignment between human and AI head movements [56].

Second, outward-facing video recordings were analysed using Google MediaPipe pose estimation. From these pose outputs, spatial engagement features were extracted, including Percentage of Time in Centre (PCT), time-in-centre duration, and spatial occupancy metrics. These measures captured embodied positioning and movement patterns within the physical theatre space, complementing XR-derived synchrony metrics.

Physiological Data. Cardiac activity was recorded using the Polar H10™ chest strap. Raw inter-beat interval (IBI) values were extracted and cleaned for artefacts [18]. Two heart rate variability indicators were computed: mean IBI, reflecting overall cardiac timing, and RMSSD (root mean square of successive differences), capturing short-term parasympathetic modulation during interaction.

Subjective Measures. Audience experience was assessed using the Narrative Engagement Scale (NES), comprising Emotional Engagement (EE), Narrative Presence (NP), Attentional Focus (AF), and Narrative Understanding (NU). Participants additionally reported a perceived conversational overlap score representing subjective alignment and an Overall Experience Rating (OER) [2, 7, 31, 47].

3.5 Data Processing and Statistical Analysis

Continuous signals were temporally synchronised and standardised (z-scores) to enable cross-modal comparison. Preprocessing included artefact removal and elimination of implausible physiological values. Associations across modalities were examined using Spearman’s rank-order correlation (ρ), selected due to non-normal

distributions and the ordinal nature of questionnaire measures. Statistical significance was assessed at $p < .05$. Analysis was structured according to the layered alignment model introduced in Section 2. Specifically, the following relationships were examined:

- Internal coherence among behavioural synchrony measures
- Associations between behavioural alignment and physiological indicators
- Relationships between perceived alignment and embodied coordination
- Links between alignment measures and experiential outcomes

Alignment was treated as a cross-layer analytical construct encompassing conversational coordination, embodied synchrony, autonomic modulation, and subjective perception.

4 Empirical Observations: Multimodal Alignment Dynamics

This section presents empirical findings structured according to the layered alignment framework introduced in Section 2. Rather than organising results solely around subjective engagement, findings are interpreted through conversational, behavioural, physiological, and perceived alignment dynamics.

4.1 Behavioural Synchrony as an Embodied Alignment Construct

Behavioural alignment was operationalised through head movement synchrony, wrist synchrony, and composite synchrony indices derived from MediaPipe-based pose tracking. Strong internal coherence was observed among these measures (Spearman ρ ranging from approximately 0.73 to 0.92), indicating that they form a stable embodied alignment construct rather than isolated movement artefacts. The high intercorrelation between head and wrist synchrony suggests that alignment manifests systemically across multiple movement channels. Increased gesture synchrony was accompanied by increased head coordination, supporting the interpretation that embodied alignment reflects coordinated interactional coupling.

4.2 Behavioural Alignment and Physiological Co-Regulation

To examine whether embodied alignment extended into autonomic regulation, associations between behavioural synchrony and inter-beat interval (IBI) were analysed. Composite behavioural synchrony measures demonstrated moderate negative correlations with mean IBI (approximately $\rho = -0.57$ to -0.62), indicating that stronger embodied coordination co-occurred with heightened physiological activation.

Lower IBI values reflect increased autonomic arousal. The observed relationship suggests that behavioural alignment is not purely kinematic but is accompanied by measurable physiological modulation, consistent with co-regulatory processes observed in human–human interaction [39].

4.3 Perceived Alignment and Experiential Outcomes

Perceived alignment, operationalised through the conversational overlap score, emerged as a central experiential predictor. Overlap score was strongly associated with EE ($\rho = 0.56$, $p < .001$) and OER ($\rho = 0.58$, $p < .001$). Participants reporting stronger conversational reciprocity consistently rated the interaction more positively. As illustrated in Figure 4, both emotional engagement and overall experience demonstrate a graded increase across overlap levels, indicating a monotonic relationship between perceived alignment and experiential evaluation.

EE further showed positive associations with Narrative Presence (NP) ($\rho = 0.42$, $p = .002$) and OER ($\rho = 0.54$, $p < .001$), positioning EE as a primary experiential mediator linking perceived alignment to overall evaluation. The interrelationships among experiential measures are illustrated in Figure 5. EE shows a strong positive association with Overall Experience Rating ($\rho = 0.67$), and a moderate association with Narrative Presence ($\rho = 0.52$). Narrative Presence similarly correlates positively with Overall Experience ($\rho = 0.52$). In contrast, Narrative Understanding and Attentional Focus exhibit inverse associations with engagement and overall evaluation, suggesting that experiential appraisal is driven more strongly by affective and immersive dimensions than by cognitive monitoring alone. Importantly, perceived alignment did not uniformly mirror embodied synchrony metrics, suggesting that subjective alignment reflects experiential appraisal but does not fully capture deeper multimodal coordination.

4.4 Alignment Trade-Offs: Attention and Narrative Monitoring

AF and NU demonstrated negative associations with both EE and overall evaluation. AF was negatively correlated with OER ($\rho = -0.32$, $p = .025$) and overlap score ($\rho = -0.53$, $p < .001$). Similarly, NU was negatively associated with EE ($\rho = -0.63$, $p < .001$) and OER ($\rho = -0.49$). These patterns indicate a potential trade-off between affective alignment and analytical monitoring. Heightened cognitive tracking of narrative detail coincided with reduced conversational reciprocity and diminished emotional immersion. Alignment may therefore involve a redistribution of attentional resources from evaluative monitoring toward relational attunement.

4.5 Narrative Presence and Temporal Exposure

NP was positively associated with session duration ($\rho = 0.29$, $p = .047$), suggesting that extended exposure facilitated deeper immersion. NP also correlated positively with overlap score ($\rho = 0.36$, $p = .025$) and OER ($\rho = 0.45$, $p = .001$), reinforcing its role as an experiential bridge between perceived alignment and overall evaluation.

4.6 Voice-Derived and Physiological Indicators

Voice-derived affective features demonstrated strong internal coherence with sentiment-based engagement measures ($\rho = 0.99$, $p < .001$), reflecting the convergent validity of vocal affective indicators. While these features were not treated as independent

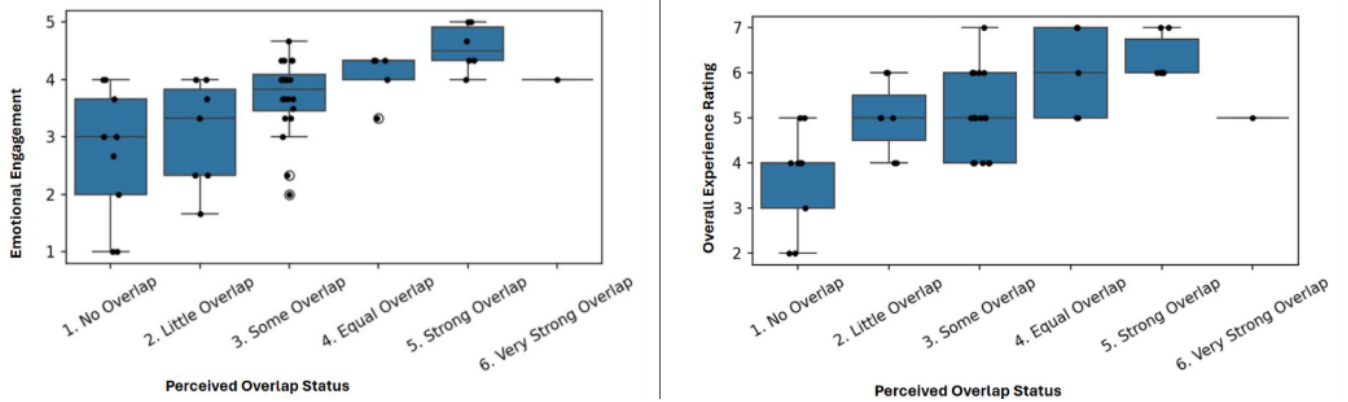


Figure 4: Emotional engagement (left) and overall experience rating (right) by perceived overlap status. Both outcomes demonstrate a monotonic increase across overlap levels, supporting the positive association between subjective alignment and experiential evaluation. Boxes represent interquartile ranges, central lines indicate medians, and points denote individual responses.

Table 1: Overview of raw multimodal signals and derived alignment metrics

Layer	Raw Data Collected	Derived Metrics	Alignment Dimension
Conversational	Single audio recording containing both participant and AI speech	Speech segmentation, turn-taking intervals, speech overlap duration, conversational pacing metrics	Conversational synchrony
Behavioral (XR)	HMD 6DoF positional and rotational heat logs	Head pitch trajectories, sliding-window coherence analysis, synchrony percentage	Embodied synchrony
Behavioral (Video)	Outward-facing video recording	MediaPipe pose estimation; Percentage of Time in Centre (PCT), time-in-centre duration, spatial occupancy features	Spatial engagement dynamics
Physiological	Raw inter-beat intervals (IBI) from Polar H10	Mean IBI, RMSSD (root mean square of successive differences)	Autonomic modulation
Subjective	Narrative Engagement Scale (EE, NP, AF, NU), perceived overlap score, Overall Experience Rating (OER)	Scale scores (EE, NP, AF, NU, Overlap, OER)	Perceived alignment & experiential evaluation

predictors, they support the interpretation that affective alignment was reflected across subjective and acoustic modalities.

Mean IBI showed a negative association with Attentional Focus ($\rho = -0.33$, $p = .037$), indicating heightened physiological activation during periods of increased monitoring. Together with behavioural findings, this supports the interpretation that alignment processes are accompanied by measurable autonomic modulation.

4.7 Integrated Predictors of Overall Evaluation

To examine the relative contribution of behavioural and experiential variables to overall experience, Spearman correlations were computed and ranked by magnitude (Figure 6). Overlap score ($\rho = 0.58$) and Emotional Engagement ($\rho = 0.54$) emerged as the strongest positive predictors of overall evaluation. Narrative Presence showed a moderate positive association, while gaze-based metrics and attentional focus exhibited inverse relationships. These findings indicate that perceived alignment and affective engagement contribute more

strongly to experiential evaluation than low-level behavioural indicators alone.

Across modalities, the strongest positive predictors of OER were perceived overlap ($\rho = 0.58$, $p < .001$), Emotional Engagement ($\rho = 0.54$, $p < .001$), and Narrative Presence ($\rho = 0.45$, $p = .001$). Negative predictors included Attentional Focus ($\rho = -0.32$, $p = .025$) and Narrative Understanding ($\rho = -0.31$, $p = .032$). Collectively, these findings indicate that overall evaluation was shaped primarily by affective immersion and perceived conversational reciprocity, rather than sustained analytical monitoring or detailed narrative comprehension. Taken together, the empirical results support a layered interpretation of bidirectional alignment: embodied synchrony forms a coherent construct; embodied alignment co-occurs with physiological modulation; perceived alignment strongly predicts experiential quality; and alignment dynamics may involve attentional trade-offs across cognitive and affective dimensions.

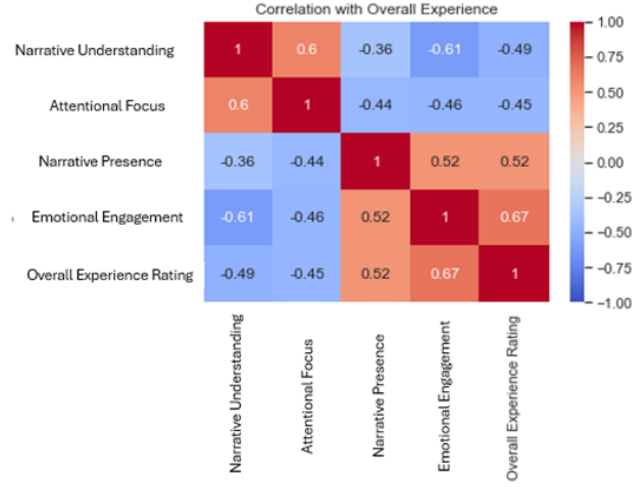


Figure 5: Correlation matrix among experiential measures. Emotional engagement and narrative presence show strong positive associations with overall experience rating, while narrative understanding and attentional focus exhibit inverse relationships, highlighting the layered structure of experiential evaluation.

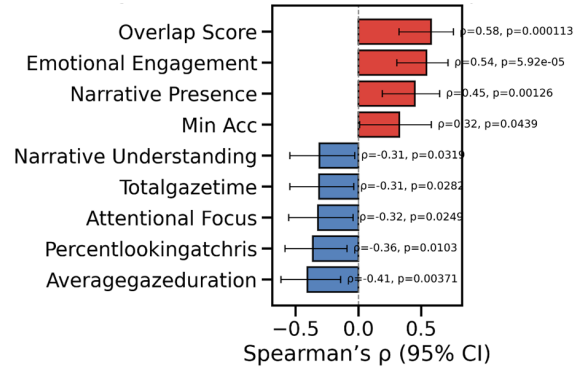


Figure 6: Spearman correlations (ρ) with 95% confidence intervals for predictors of overall experience rating. Positive associations indicate variables linked to higher experiential evaluation, while negative associations indicate inverse relationships. Overlap score and emotional engagement emerge as the strongest positive predictors.

5 Discussion: Implications for Bidirectional Alignment Evaluation and Design

The findings extend the notion of human-AI alignment beyond model-level objective matching toward a dynamic process of reciprocal adaptation unfolding within social collaboration with virtual agents [1, 46]. Across conversational, behavioural, physiological, and subjective dimensions, alignment emerged as a multilayered construct that cannot be reduced to a single metric. Importantly, subjective experiential measures—particularly Emotional Engagement, Narrative Presence, perceived overlap, and Overall Experience Rating (OER)—formed the foundational evaluative layer upon which other alignment indicators were interpreted.

Table 2: Spearman correlations between predictors and outcome measures.

Predictor	Outcome	ρ	p
Narrative Presence	Emotional Engagement	0.42	0.002
Narrative Understanding	Emotional Engagement	-0.63	<0.001
Overall Exp Rating	Emotional Engagement	0.54	<0.001
Overlap Score	Emotional Engagement	0.56	<0.001
Mean IBI	Emotional Engagement	-0.35	0.028
% Gaze on AI	Attentional Focus	0.33	0.037
Overlap Score	Overall Experience	0.58	<0.001
Emotional Engagement	Overall Experience	0.54	<0.001
Narrative Presence	Overall Experience	0.45	0.001
Min Acc	Overall Experience	0.32	0.044

Empirically, OER functioned as the integrative endpoint of the alignment process. Perceived conversational reciprocity showed strong associations with OER, indicating that subjective appraisal anchors the experiential meaning of collaboration. On this experiential layer, additional forms of synchrony were layered: conversational timing dynamics, embodied movement synchrony derived from XR heat logs, and physiological modulation (mean IBI and RMSSD). Behavioural synchrony measures formed a coherent embodied construct, and their association with autonomic regulation suggests cross-layer co-regulation [12, 39]. As shown in Table 2, subjective and synchrony-based predictors demonstrate differential associations with OER and related experiential outcomes.

This layered structure supports a re-conceptualisation of alignment as interdependent but asymmetrical across levels. Subjective perception provides the experiential grounding of collaboration, while conversational, behavioural, and physiological synchrony offer complementary insights into how coordination unfolds beneath conscious awareness. Evaluating alignment solely through perceived responsiveness risks overlooking embodied co-regulation, whereas focusing exclusively on technical synchrony metrics may fail to capture user experience [32, 43]. A bidirectional evaluation framework must therefore integrate phenomenological and computational indicators in parallel.

The negative associations observed between Attentional Focus, Narrative Understanding, and affective engagement further illuminate this layered dynamic. Increased analytical monitoring coincided with reduced emotional engagement and lower OER, suggesting redistribution of cognitive resources during interaction [7, 47]. These findings indicate that alignment in embodied human-AI systems may be facilitated more by affective attunement than by intensified cognitive scrutiny.

Taken together, the results motivate three principles for evaluating bidirectional alignment:

- **Experiential anchoring:** Subjective measures, particularly OER and perceived reciprocity, should serve as the foundational layer for interpreting alignment.
- **Multimodal integration:** Conversational synchrony, movement synchrony from XR heat logs, and physiological indicators should be layered onto subjective evaluation to capture deeper coordination processes [33, 34].
- **Cross-layer coherence:** Evaluation should assess whether synchrony indicators converge across layers, signalling robust reciprocal adaptation.

Beyond evaluation, these findings inform adaptive system design. If alignment unfolds through coordinated conversational timing and embodied synchrony, virtual agents may dynamically adjust pacing, gesture expressiveness, or response latency to sustain reciprocal attunement [1, 8, 25, 26]. Crucially, such adaptation must preserve human agency rather than override user intent [46].

Overall, the results demonstrate that bidirectional human-AI alignment in social collaboration emerges through layered coordination processes anchored in subjective experience and enriched by conversational, behavioural, and physiological synchrony. By structuring evaluation around an experiential foundation while integrating multimodal indicators, this work contributes a methodological framework for assessing and designing value-centred, co-evolving human-AI partnerships.

6 Limitations and Future Work.

While the proposed framework provides a structured approach for evaluating bidirectional human-AI alignment across conversational, behavioural, physiological, and perceived layers, several limitations should be noted. The empirical evaluation was conducted in a specific XR narrative interaction scenario with a single AI character, which may limit the generalisability of the findings to other contexts, such as collaborative task environments, multi-agent systems, or long-term human-AI partnerships. In addition, although perceived alignment was measured through subjective overlap and engagement scales, subjective interpretations of alignment are inherently shaped by contextual and individual factors. Prior work suggests that elements such as VR fatigue, prior familiarity with immersive systems, the topic of interaction, and participants' interests, motivations, or expectations may influence how alignment is experienced during human-AI interaction [37, 48]. Future research could extend the framework by explicitly modelling these contextual and experiential factors, enabling a more dynamic understanding of how perceived alignment emerges during embodied interaction. Longitudinal studies examining repeated interactions with adaptive AI agents may also provide deeper insight into how conversational coordination, behavioural synchrony, physiological modulation, and subjective experience co-evolve over time.

7 Conclusion

This paper reframes human-AI alignment within social collaboration with virtual agents as a dynamic, multimodal phenomenon rather than a static model property. We operationalised alignment as emerging through coordinated interaction across conversational timing, embodied movement synchrony derived from XR heat logs, physiological modulation, and subjective appraisal. In doing so, alignment was treated as a measurable interactional state unfolding across complementary synchrony mechanisms. Empirical results demonstrated that different forms of synchrony contribute in distinct ways to collaborative experience. Conversational synchrony—captured through turn-taking and speech overlap—strongly predicted Overall Experience Rating (OER), positioning perceived reciprocity as central to experiential evaluation. Movement synchrony derived from HMD heat logs formed a coherent behavioural construct, reflecting embodied coordination between participant and virtual agent. Physiological modulation (mean IBI

and RMSSD) further indicated that autonomic regulation co-varied with behavioural synchrony, revealing deeper alignment processes beyond conscious appraisal. Importantly, subjective alignment did not uniformly mirror embodied synchrony metrics, underscoring the layered nature of multimodal coordination in social human-AI collaboration. While affective immersion and narrative presence positively shaped OER, increased attentional monitoring and analytical processing were associated with reduced relational attunement, highlighting trade-offs between cognitive control and experiential engagement.

To the best of the authors' knowledge, this is the first study to systematically evaluate social collaboration with a virtual agent using an integrated multimodal alignment framework that combines conversational synchrony, movement synchrony from XR heat logs, physiological indicators, subjective measures, and performance-linked experiential outcomes. By linking multimodal synchrony processes directly to experiential and evaluative outcomes, this work provides a structured methodological foundation for assessing reciprocal human-AI coordination in immersive settings. Such a framework supports the design of socially adaptive virtual agents that can sustain meaningful, balanced collaboration over time.

Acknowledgments

This research was supported by the Technological University of the Shannon, Athlone, Ireland, in collaboration with the Science Foundation Ireland CONFIRM Centre (Grant No. 16/RC/3918). The work was also partially supported by the Horizon Europe TRANSMIXR project (Grant No. 101070109). Additional support was provided by Research Ireland and the Department of Agriculture, Food and the Marine on behalf of the Government of Ireland under the VistaMilk programme (Grant No. 21/RC/10303_P2).

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Received 20 March 2026; revised 25 March 2026; accepted 01 April 2026